

## 4.2 Annihilators

Section 1 built the dual space  $V^*$ , the space of linear measurements on  $V$ . Now we connect subspaces of  $V$  with subspaces of  $V^*$ . A plane in  $\mathbb{R}^3$  can be described by a spanning list or by an equation, and an equation is a functional. This section collects all the equations that a set of vectors satisfies into one subspace of  $V^*$ , the annihilator, counts its dimension, and shows that it turns sums into intersections and intersections into sums.

### 4.2.1 The equations a subspace satisfies

Every object comes with its subobjects. The dual space  $V^*$  has subspaces of its own, and the question is which of them are attached to subspaces of  $V$ . The standard example shows the way.

Let  $U = \text{span}((1, 0, 1), (0, 1, 1))$ , a plane in  $\mathbb{R}^3$ . The functional  $\varphi(x, y, z) = x + y - z$  vanishes on both spanning vectors, so by linearity it vanishes on all of  $U$ : the plane satisfies the equation  $x + y - z = 0$ . So do  $2x + 2y - 2z = 0$  and the trivial equation  $0 = 0$ . Which functionals vanish on  $U$ ? By Theorem 4.1.1 a functional is  $\psi(x, y, z) = ax + by + cz$ . It vanishes on the two spanning vectors if and only if  $a + c = 0$  and  $b + c = 0$ , that is,  $\psi = -c(x + y - z) = -c\varphi$ . So the equations of the plane form the **line**  $\text{span}(\varphi)$  in  $(\mathbb{R}^3)^*$ .

For the  $z$ -axis  $\text{span}((0, 0, 1))$ , the same computation gives  $c = 0$ : the equations are  $ax + by = 0$ , a **plane** in  $(\mathbb{R}^3)^*$ . A bigger subspace satisfies fewer equations, and in both cases the dimensions add up to 3. The set of equations deserves a name, and the pattern deserves a theorem.

*The annihilator of a set of vectors is the space of all linear equations that every one of those vectors satisfies.*

#### Definition 4.2.1 — Annihilator

Let  $V$  be a vector space over  $F$ , and let  $S \subseteq V$  be **any subset**. The **annihilator** of  $S$  is

$$S^0 := \{\varphi \in V^* : \varphi(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}.$$

In words:  $S^0$  is a subset of the **dual space**  $V^*$ , not of  $V$ . A functional belongs to it when it vanishes at **every** vector of  $S$ ; a single non-zero value disqualifies it. The set  $S$  needs no structure: it may be a subspace, a finite list of vectors, or even empty. The superscript 0 recalls the value that the functionals take on  $S$ .

**Well-definedness.** The definition makes sense for every subset. What needs checking is that the result is a subspace, which is what makes its dimension meaningful.

#### Proposition 4.2.1 — The Annihilator Is a Subspace

Let  $V$  be a vector space over  $F$ , and let  $S \subseteq V$ . Then  $S^0$  is a subspace of  $V^*$ .

*Proof.* We use the subspace test (Theorem 1.2.1) in  $V^*$ . The zero functional sends every  $\mathbf{s} \in S$  to 0, so it lies in  $S^0$ . Let  $\varphi, \psi \in S^0$  and  $c \in F$ . For every  $\mathbf{s} \in S$ , the pointwise

operations of Definition 4.1.2 give

$$(\varphi + \psi)(\mathbf{s}) = \varphi(\mathbf{s}) + \psi(\mathbf{s}) = 0 + 0 = 0, \quad (c\varphi)(\mathbf{s}) = c\varphi(\mathbf{s}) = c \cdot 0 = 0.$$

Hence  $\varphi + \psi \in S^0$  and  $c\varphi \in S^0$ , and  $S^0$  is a subspace of  $V^*$ . ■

### Example. (First Annihilators)

- (a) For any vector space  $V$ , find  $\{\mathbf{0}\}^0$ ,  $\emptyset^0$  and  $V^0$ .
- (b) In  $M_2(F)$ , find the annihilator of the single matrix  $E_{11}$ .
- (c) In  $\mathbb{R}[x]_{\leq 2}$ , find the annihilator of  $S = \{1, x\}$ .

*Solution.*

- (a) Every functional sends  $\mathbf{0}$  to 0 (Theorem 3.1.1), so  $\{\mathbf{0}\}^0 = V^*$ . The condition “for all  $\mathbf{s} \in \emptyset$ ” holds vacuously, so  $\emptyset^0 = V^*$ . A functional in  $V^0$  takes the value 0 at every vector, so it is the zero functional, and  $V^0 = \{0\}$ . These degenerate cases fix the direction of everything that follows: the smallest subspace of  $V$  has the largest annihilator, and the largest has the smallest.
- (b) Let  $\varepsilon_{ij}(A) = a_{ij}$  be the dual basis of the standard basis  $(E_{11}, E_{12}, E_{21}, E_{22})$ . By Theorem 4.1.3 (c), every  $\psi \in M_2(F)^*$  is  $\psi = \sum_{i,j} \psi(E_{ij}) \varepsilon_{ij}$ . Such a  $\psi$  lies in  $\{E_{11}\}^0$  exactly when  $\psi(E_{11}) = 0$ . Hence

$$\{E_{11}\}^0 = \text{span}(\varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22}),$$

the functionals  $A \mapsto c_{12}a_{12} + c_{21}a_{21} + c_{22}a_{22}$ . It has dimension 3, and  $3 + 1 = 4 = \dim M_2(F)$ .

- (c) Let  $(\varphi_0, \varphi_1, \varphi_2)$  be the dual basis of  $(1, x, x^2)$ , so  $\varphi_k$  reads off the coefficient of  $x^k$ . A functional  $\psi = \psi(1)\varphi_0 + \psi(x)\varphi_1 + \psi(x^2)\varphi_2$  lies in  $S^0$  exactly when  $\psi(1) = \psi(x) = 0$ , that is, when  $\psi = \psi(x^2)\varphi_2$ . So  $S^0 = \text{span}(\varphi_2)$ : the only equations satisfied by both 1 and  $x$  are the multiples of “the coefficient of  $x^2$  is 0”.

**Non-example by minimal change.** Shift the plane of the opening example off the origin:  $A = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 1\}$ . The set is still cut out by the same left-hand side, but  $\varphi(x, y, z) = x + y - z$  takes the value 1, not 0, on  $A$ , so the defining clause  $\varphi(\mathbf{s}) = 0$  fails and  $\varphi \notin A^0$ . In fact  $A^0 = \{0\}$ : the set  $A$  contains  $(1, 0, 0)$ ,  $(0, 1, 0)$  and  $(0, 0, -1)$ , and a functional  $ax + by + cz$  vanishing at these three vectors has  $a = b = c = 0$ . Annihilators record only **homogeneous** equations.

**Why this definition.** We allow any subset  $S$ , not only subspaces, because subspaces are usually handed to us by spanning lists, and we want to test a functional on the list alone. The theorem at the end of this section shows that this loses nothing:  $S^0 = (\text{span } S)^0$ . The name is literal. To annihilate is to reduce to nothing, and the functionals in  $S^0$  reduce every vector of  $S$  to 0.

**Warning.**  $U^0$  lives in  $V^*$ , and it is not a complement of  $U$ . It is tempting to turn the functional  $ax + by$  into the vector  $(a, b)$  and to picture  $U^0$  as a subspace of  $F^2$  “perpendicular” to  $U$ . Over  $\mathbb{F}_2$  this picture fails completely. Let  $U = \text{span}((1, 1)) \subseteq \mathbb{F}_2^2$ .

A functional  $ax + by$  vanishes on  $(1, 1)$  exactly when  $a + b = 0$ , that is,  $a = -b$ , so  $U^0 = \{0, x + y\}$ . The functional  $x + y$  corresponds to the vector  $(1, 1)$ , which lies **in**  $U$ . So the “picture” of  $U^0$  is  $U$  itself, not a complement. Perpendicularity needs an inner product, which is the orthogonal complement of Chapter 10; the annihilator needs nothing.

The payoff of the definition is the pattern we saw in the opening example. The equations of a subspace take up exactly the dimensions that the subspace leaves free.

## 4.2.2 Counting equations

### Theorem 4.2.2 — Dimension of the Annihilator

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $U$  be a subspace of  $V$ . Then

$$\dim U + \dim U^0 = \dim V.$$

More precisely, if  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$  is a basis of  $V$  extending a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  of  $U$ , with dual basis  $(\varphi_1, \dots, \varphi_n)$ , then  $(\varphi_{m+1}, \dots, \varphi_n)$  is a basis of  $U^0$ .

*Idea.* This is a dimension formula, so we start from the smallest subspace,  $U$ , take a basis and extend it to  $V$ . The dual basis now splits in two. The last  $n - m$  functionals vanish on every  $\mathbf{u}_i$ , by the delta, so they vanish on  $U$ . The first  $m$  do not. To see that nothing else is in  $U^0$ , expand a functional  $\psi \in U^0$  in the dual basis: its coefficients are its values on the basis, and the first  $m$  values are  $\psi(\mathbf{u}_i) = 0$ .

*Proof.* Since  $U$  is a subspace of the finite-dimensional space  $V$ , it has a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  (Theorem 1.6.7, Corollary 1.5.5), and by the Basis Extension Theorem (Theorem 1.6.5) it extends to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$  of  $V$ , where  $n = \dim V$ . Let  $(\varphi_1, \dots, \varphi_n)$  be the dual basis of  $\mathcal{B}$  (Theorem 4.1.3).

*The  $\varphi_j$  with  $j > m$  lie in  $U^0$ .* Let  $j > m$  and  $\mathbf{u} \in U$ , say  $\mathbf{u} = a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m$ . By Theorem 3.1.4,  $\varphi_j(\mathbf{u}) = \sum_{i=1}^m a_i\varphi_j(\mathbf{u}_i) = 0$ , since  $\varphi_j(\mathbf{u}_i) = \delta_{ji} = 0$  for  $i \leq m < j$ .

*They span  $U^0$ .* Let  $\psi \in U^0$ . By Theorem 4.1.3 (c),

$$\psi = \sum_{i=1}^m \psi(\mathbf{u}_i) \varphi_i + \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j = \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j,$$

because  $\psi(\mathbf{u}_i) = 0$  for  $\mathbf{u}_i \in U$ . Hence  $U^0 \subseteq \text{span}(\varphi_{m+1}, \dots, \varphi_n)$ , and the reverse inclusion holds because  $U^0$  is a subspace (Proposition 4.2.1, Theorem 1.3.1).

*They are independent,* being part of the basis  $(\varphi_1, \dots, \varphi_n)$  of  $V^*$ . Therefore  $(\varphi_{m+1}, \dots, \varphi_n)$  is a basis of  $U^0$ , and  $\dim U^0 = n - m = \dim V - \dim U$ . This proves the theorem. ■

Alternatively, restrict functionals to  $U$ . The map  $R: V^* \rightarrow U^*$ ,  $\varphi \mapsto \varphi|_U$ , is linear, and its kernel is  $U^0$  by definition. It is surjective: given  $\xi \in U^*$ , Theorem 3.1.5 gives  $\varphi \in V^*$  with  $\varphi(\mathbf{u}_i) = \xi(\mathbf{u}_i)$  and  $\varphi(\mathbf{v}_j) = 0$ , and then  $\varphi|_U = \xi$  because both agree on a basis of  $U$ . By the Rank–

Nullity Theorem (Theorem 3.3.1) and Corollary 4.1.4,  $\dim V = \dim V^* = \dim U^0 + \dim U^* = \dim U^0 + \dim U$ . This route shows **why** the formula holds: a functional on  $V$  is a functional on  $U$  plus a choice of values off  $U$ , and  $U^0$  measures that choice.

The theorem is the second signature move of the chapter: **pass to the dual and count**. One immediate consequence: if  $U$  is a **proper** subspace of a finite-dimensional  $V$ , then  $\dim U^0 > 0$ , so some **non-zero** functional vanishes on  $U$ . Equivalently, a list spans a finite-dimensional  $V$  if and only if the only functional that kills every vector of the list is 0: apply the previous sentence to the span  $U$  of the list, and note that a functional kills every vector of the list exactly when it kills  $U$ , by Theorem 3.1.4.

In  $F^n$  the whole computation becomes elimination, because functionals are rows.

### Example. (An Annihilator as a Null Space)

Find a basis of  $U^0$  for  $U = \text{span}((1, 2, 0), (0, 1, 1)) \subseteq \mathbb{R}^3$ , and check the dimension formula.

*Solution.* By Theorem 4.1.1, every functional is  $\varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$  for a unique  $\mathbf{a} = (a_1, a_2, a_3)$ . By linearity,  $\varphi_{\mathbf{a}}$  vanishes on  $U$  if and only if it vanishes on the two spanning vectors (a combination of vectors killed by  $\varphi_{\mathbf{a}}$  is killed, by Theorem 3.1.4). These two conditions are

$$a_1 + 2a_2 = 0, \quad a_2 + a_3 = 0, \quad \text{that is,} \quad A\mathbf{a} = \mathbf{0} \quad \text{with} \quad A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

the matrix whose **rows** are the spanning vectors. So  $\varphi_{\mathbf{a}} \in U^0$  if and only if  $\mathbf{a} \in \text{nul}(A)$ . Subtracting twice row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix},$$

with  $a_3$  free. By Theorem 2.5.4,  $\text{nul}(A) = \text{span}((2, -1, 1))$ . Hence  $U^0 = \text{span}(\psi)$  with

$$\psi(x, y, z) = 2x - y + z.$$

Check:  $\psi(1, 2, 0) = 2 - 2 + 0 = 0$  and  $\psi(0, 1, 1) = -1 + 1 = 0$ . The two spanning vectors are independent (neither is a multiple of the other), so  $\dim U = 2$ , and  $\dim U + \dim U^0 = 2 + 1 = 3$ , as Theorem 4.2.2 predicts. As a by-product,  $U$  is the plane  $2x - y + z = 0$ :  $U \subseteq \ker \psi$ , both have dimension 2 (Proposition 4.1.2), and Theorem 1.6.8 gives equality.

The same recipe works in general. For  $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$ , let  $A$  have rows  $\mathbf{u}_1^T, \dots, \mathbf{u}_m^T$ . Then  $U^0 = \{\varphi_{\mathbf{a}} : \mathbf{a} \in \text{nul}(A)\}$ , and the isomorphism of Theorem 4.1.1 carries a basis of  $\text{nul}(A)$  to a basis of  $U^0$ .

Away from  $F^n$ , counting often replaces the computation entirely.

### Example. (The Equations of a Root Condition)

Let  $U = \{p \in \mathbb{R}[x]_{\leq 2} : p(1) = 0\}$ . Find  $U^0$ .

*Solution.* Let  $\varepsilon_1(p) = p(1)$ , a linear functional (Example (Four Families of Functionals)). By definition  $U = \ker \varepsilon_1$ , so  $\varepsilon_1 \in U^0$ . Since  $\varepsilon_1(1) = 1$ ,  $\varepsilon_1 \neq 0$ , and Proposition 4.1.2

gives  $\dim U = 3 - 1 = 2$ . By Theorem 4.2.2,  $\dim U^0 = 3 - 2 = 1$ . Now  $\text{span}(\varepsilon_1)$  is a subspace of  $U^0$  (Proposition 4.2.1) of dimension 1, so  $U^0 = \text{span}(\varepsilon_1)$  by Theorem 1.6.8.

We never found a basis of  $U$  and never solved an equation. The conclusion is concrete: a functional on  $\mathbb{R}[x]_{\leq 2}$  vanishes on every polynomial with a root at 1 **only if** it is a multiple of  $p \mapsto p(1)$ .

*Quick check.* Let  $U = \text{span}((1, 1, 1)) \subseteq \mathbb{R}^3$ . What is  $\dim U^0$ ? Name a basis of  $U^0$  without solving a system.

*Solution.*  $\dim U^0 = 3 - 1 = 2$  by Theorem 4.2.2. The functionals  $x - y$  and  $y - z$  vanish at  $(1, 1, 1)$ , hence on  $U$ , and they are independent: in  $a(x - y) + b(y - z) = 0$ , the coefficient of  $x$  gives  $a = 0$ , and then  $b = 0$ . Two independent vectors in a 2-dimensional space form a basis (Theorem 1.6.4).

### 4.2.3 Sums, intersections and spans

Taking annihilators reverses inclusions: more vectors satisfy fewer common equations. It also exchanges sums with intersections. In  $\mathbb{R}^3$ , let  $U$  be the  $xy$ -plane and  $W$  the  $xz$ -plane. Then  $U^0 = \text{span}(z)$  and  $W^0 = \text{span}(y)$ , writing  $x, y, z$  for the coordinate functionals. The intersection  $U \cap W$  is the  $x$ -axis, with annihilator  $\text{span}(y, z) = U^0 + W^0$ . The sum  $U + W = \mathbb{R}^3$  has annihilator  $\{0\} = U^0 \cap W^0$ . The theorem says this always happens.

#### Theorem 4.2.3 — Properties of Annihilators

Let  $V$  be a vector space over  $F$ .

- (a) If  $S \subseteq T \subseteq V$ , then  $T^0 \subseteq S^0$ .
- (b) For every subset  $S \subseteq V$ ,  $S^0 = (\text{span } S)^0$ .
- (c) For subspaces  $U, W$  of  $V$ ,  $(U + W)^0 = U^0 \cap W^0$ .
- (d) If  $V$  is **finite-dimensional**, then for subspaces  $U, W$  of  $V$ ,  $(U \cap W)^0 = U^0 + W^0$ .

*Idea.* Parts (a)–(c) are unwinding definitions: fix a functional in one side and test it on the vectors of the other. In (d), the inclusion  $U^0 + W^0 \subseteq (U \cap W)^0$  is also direct. The reverse inclusion is the hard half. Given  $\varphi$  vanishing on  $U \cap W$ , we would have to **split** it as  $\psi + \chi$  with  $\psi$  killing  $U$  and  $\chi$  killing  $W$ , and nothing in sight tells us how. So we count instead: one inclusion plus equal dimensions gives equality. The plan is ① compute  $\dim(U^0 + W^0)$  by the dimension formula for sums inside  $V^*$ ; ② replace  $U^0 \cap W^0$  by  $(U + W)^0$  using (c); ③ turn every annihilator dimension into a dimension in  $V$  with Theorem 4.2.2; ④ recognize the dimension formula in  $V$ . Finite dimension is spent in step ③.

*Proof.*

- (a) Let  $\varphi \in T^0$ . Every  $\mathbf{s} \in S$  lies in  $T$ , so  $\varphi(\mathbf{s}) = 0$ . Hence  $\varphi \in S^0$ .
- (b) Since  $S \subseteq \text{span } S$  (Theorem 1.3.1), part (a) gives  $(\text{span } S)^0 \subseteq S^0$ . Conversely, let  $\varphi \in S^0$  and  $\mathbf{v} \in \text{span } S$ , say  $\mathbf{v} = a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k$  with  $\mathbf{s}_i \in S$  and  $k \geq 0$ . By

Theorem 3.1.4,  $\varphi(\mathbf{v}) = \sum_i a_i \varphi(\mathbf{s}_i) = 0$ . Hence  $\varphi \in (\text{span } S)^0$ .

(c) Since  $U \subseteq U+W$  and  $W \subseteq U+W$  (Theorem 1.7.1), part (a) gives  $(U+W)^0 \subseteq U^0$  and  $(U+W)^0 \subseteq W^0$ , so  $(U+W)^0 \subseteq U^0 \cap W^0$ . Conversely, let  $\varphi \in U^0 \cap W^0$ , and let  $\mathbf{u} + \mathbf{w} \in U+W$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ . Then  $\varphi(\mathbf{u} + \mathbf{w}) = \varphi(\mathbf{u}) + \varphi(\mathbf{w}) = 0 + 0 = 0$ . Hence  $\varphi \in (U+W)^0$ .

(d)  $(\supseteq)$  Let  $\varphi = \psi + \chi$  with  $\psi \in U^0$  and  $\chi \in W^0$ , and let  $\mathbf{v} \in U \cap W$ . Since  $\mathbf{v} \in U$  and  $\mathbf{v} \in W$ ,  $\varphi(\mathbf{v}) = \psi(\mathbf{v}) + \chi(\mathbf{v}) = 0 + 0 = 0$ . Hence  $U^0 + W^0 \subseteq (U \cap W)^0$ .

$(\subseteq)$  Let  $n = \dim V$ . The subspaces  $U$ ,  $W$ ,  $U \cap W$  (Theorem 1.2.2) and  $U+W$  (Theorem 1.7.1) of  $V$  are finite-dimensional by Theorem 1.6.7. The space  $V^*$  has dimension  $n$  by Corollary 4.1.4, so its subspaces  $U^0$ ,  $W^0$  are finite-dimensional too. By the Dimension Formula for Sums (Theorem 1.7.3) in  $V^*$ , then (c),

$$\dim(U^0 + W^0) = \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) = \dim U^0 + \dim W^0 - \dim(U \cap W)^0.$$

By Theorem 4.2.2 applied to  $U$ ,  $W$  and  $U+W$ , the right-hand side is

$$(n - \dim U) + (n - \dim W) - (n - \dim(U+W)) = n - (\dim U + \dim W - \dim(U+W)) = n - \dim(U \cap W),$$

where the last equality is Theorem 1.7.3 in  $V$ . By Theorem 4.2.2 once more,  $n - \dim(U \cap W) = \dim(U \cap W)^0$ . So  $U^0 + W^0$  is a subspace of  $(U \cap W)^0$  (by  $(\supseteq)$  and Theorem 1.7.1) with the same finite dimension, and Theorem 1.6.8 gives  $U^0 + W^0 = (U \cap W)^0$ . This proves the theorem. ■

Part (b) is what made the computations of this section legitimate: to find the annihilator of a subspace, test functionals on a spanning list only. Parts (c) and (d) form a dictionary. A subspace given as a **sum** has equations obtained by **intersecting** equation spaces; a subspace given as an **intersection** has equations obtained by **adding** them. Later in this chapter this dictionary becomes a practical method for computing sums and intersections of subspaces.

*Remark.* Parts (a)–(c) and the inclusion  $(\supseteq)$  in (d) hold in every vector space; only the counting step used finite dimension. The equality (d) is in fact true for **every**  $V$ , but the proof must then construct the splitting  $\varphi = \psi + \chi$  by hand; Exercise C1 below does this. In finite dimension, (a) also becomes strict: if  $U \subsetneq W$ , then  $\dim U < \dim W$  (Theorem 1.6.8), so  $\dim W^0 < \dim U^0$  and  $W^0 \subsetneq U^0$ .

## 4.2.4 Exercises

### 4.2.4.1 A. Check your understanding

#### Exercise A1.

- (a) Define the annihilator  $S^0$  of a subset  $S$  of a vector space  $V$ .
- (b) True or false:  $U^0$  is a subspace of  $V$ . Justify your answer.
- (c) Let  $U$  be a 2-dimensional subspace of  $\mathbb{R}^5$ . What is  $\dim U^0$ ?
- (d) For a finite-dimensional  $V$ , what are  $\{\mathbf{0}\}^0$  and  $V^0$ ?
- (e) True or false: if  $U \subsetneq W$  are subspaces of a finite-dimensional  $V$ , then  $W^0 \subsetneq U^0$ .

Justify your answer.

- (f) To compute  $U^0$  for  $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$ , which matrix do you row reduce, and why?

*Solution.*

- (a)  $S^0 = \{\varphi \in V^* : \varphi(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}$  (Definition 4.2.1).  
 (b) False.  $U^0$  is a subspace of the dual space  $V^*$  (Proposition 4.2.1); its elements are functionals, not vectors of  $V$ .  
 (c)  $\dim U^0 = 5 - 2 = 3$ , by Theorem 4.2.2.  
 (d)  $\{\mathbf{0}\}^0 = V^*$  and  $V^0 = \{0\}$ , by Example (First Annihilators) (a).  
 (e) True. By Theorem 4.2.3 (a),  $W^0 \subseteq U^0$ . Since  $U \subsetneq W$ ,  $\dim U < \dim W$  by Theorem 1.6.8, so  $\dim W^0 = \dim V - \dim W < \dim V - \dim U = \dim U^0$ . Hence the inclusion is strict.  
 (f) The matrix  $A$  whose rows are  $\mathbf{u}_1^T, \dots, \mathbf{u}_m^T$ . By Theorem 4.1.1 a functional is  $\mathbf{x} \mapsto \mathbf{a}^T \mathbf{x}$ , and it kills every  $\mathbf{u}_i$  exactly when  $A\mathbf{a} = \mathbf{0}$ ; so  $U^0$  corresponds to  $\text{nul}(A)$  (Example (An Annihilator as a Null Space)).

#### 4.2.4.2 B. Practice

##### Exercise B1: Computing annihilators.

- (a) Let  $U = \text{span}((1, 2, 1, 0), (2, 4, 3, 1), (1, 2, 2, 1)) \subseteq \mathbb{R}^4$ . Find a basis of  $U^0$ , and verify that  $\dim U + \dim U^0 = 4$ .  
 (b) Let  $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = p(-1) = 0\}$ . Show that  $U^0 = \text{span}(\varepsilon_1, \varepsilon_{-1})$ , where  $\varepsilon_c(p) = p(c)$ .

*Solution.*

- (a) Let  $A$  have the three spanning vectors as rows. As in Example (An Annihilator as a Null Space),  $\varphi_{\mathbf{a}} \in U^0$  if and only if  $A\mathbf{a} = \mathbf{0}$ . Subtracting 2 times row 1 from row 2 and row 1 from row 3 gives rows  $(1, 2, 1, 0)$ ,  $(0, 0, 1, 1)$ ,  $(0, 0, 1, 1)$ ; subtracting row 2 from row 3 and then row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The free variables are  $a_2$  and  $a_4$ , and Theorem 2.5.4 gives  $\text{nul}(A) = \text{span}((-2, 1, 0, 0), (1, 0, -1, 1))$ . Hence  $U^0$  has the basis

$$\psi_1(x_1, x_2, x_3, x_4) = -2x_1 + x_2, \quad \psi_2(x_1, x_2, x_3, x_4) = x_1 - x_3 + x_4.$$

Check on the first two spanning vectors:  $\psi_1(1, 2, 1, 0) = 0$ ,  $\psi_2(1, 2, 1, 0) = 0$ ,  $\psi_1(2, 4, 3, 1) = 0$ ,  $\psi_2(2, 4, 3, 1) = 2 - 3 + 1 = 0$ ; the third vector is their difference,  $(2, 4, 3, 1) - (1, 2, 1, 0)$ . The RREF has two pivots, so  $\dim U = \dim \text{row}(A) = 2$  (Theorem 2.5.5), and  $2 + 2 = 4$ .

- (b) By definition  $\varepsilon_1, \varepsilon_{-1} \in U^0$ . *Independent:* if  $a\varepsilon_1 + b\varepsilon_{-1} = 0$ , applying it to  $x + 1$  gives  $2a = 0$  and applying it to  $x - 1$  gives  $-2b = 0$ . *Dimension of  $U$ :* the map

$T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$ ,  $p \mapsto (p(1), p(-1))$ , is linear with kernel  $U$ , and its image contains  $T(x+1) = (2, 0)$  and  $T(x-1) = (0, -2)$ , which span  $\mathbb{R}^2$ . So  $\text{rank } T = 2$  and  $\dim U = 4 - 2 = 2$  by Theorem 3.3.1. By Theorem 4.2.2,  $\dim U^0 = 2$ . Hence the 2-dimensional subspace  $\text{span}(\varepsilon_1, \varepsilon_{-1})$  of  $U^0$  is all of  $U^0$  (Theorem 1.6.8).

### Exercise B2: Equations of the symmetric matrices.

Let  $U = \{A \in M_2(\mathbb{R}) : A^T = A\}$ . Find  $U^0$ , and verify the dimension formula.

*Solution.* Every symmetric matrix is  $aE_{11} + b(E_{12} + E_{21}) + cE_{22}$ , so  $U = \text{span}(E_{11}, E_{12} + E_{21}, E_{22})$ . Let  $(\varepsilon_{ij})$  be the dual basis of the standard basis, so that every  $\psi \in M_2(\mathbb{R})^*$  is  $\psi = \sum_{i,j} c_{ij}\varepsilon_{ij}$  with  $c_{ij} = \psi(E_{ij})$  (Theorem 4.1.3 (c)). By Theorem 4.2.3 (b),  $\psi \in U^0$  if and only if  $\psi$  kills the three spanning matrices:

$$c_{11} = 0, \quad c_{12} + c_{21} = 0, \quad c_{22} = 0.$$

So  $\psi = c_{12}(\varepsilon_{12} - \varepsilon_{21})$ , and  $U^0 = \text{span}(\varepsilon_{12} - \varepsilon_{21})$ : the only equation of the symmetric matrices, up to scaling, is  $a_{12} - a_{21} = 0$ . The three spanning matrices are independent (they involve disjoint sets of entries), so  $\dim U = 3$ , and  $\dim U + \dim U^0 = 3 + 1 = 4 = \dim M_2(\mathbb{R})$ .

### Exercise B3: The equations of a hyperplane.

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\varphi \in V^*$  be non-zero. Prove that  $(\ker \varphi)^0 = \text{span}(\varphi)$ .

*Solution.* By definition of the kernel,  $\varphi$  vanishes on  $\ker \varphi$ , so  $\varphi \in (\ker \varphi)^0$ , and  $\text{span}(\varphi) \subseteq (\ker \varphi)^0$  by Proposition 4.2.1. Since  $\varphi \neq 0$ ,  $\dim \text{span}(\varphi) = 1$ , and Proposition 4.1.2 gives  $\dim \ker \varphi = \dim V - 1$ . By Theorem 4.2.2,  $\dim(\ker \varphi)^0 = \dim V - (\dim V - 1) = 1$ . A 1-dimensional subspace of a 1-dimensional space is the whole space (Theorem 1.6.8), so  $(\ker \varphi)^0 = \text{span}(\varphi)$ .

#### 4.2.4.3 C. Going deeper

### Exercise C1: The intersection formula without counting.

Let  $U, W$  be subspaces of a vector space  $V$  over  $F$ , and let  $\varphi \in (U \cap W)^0$ .

- Suppose  $V$  is finite-dimensional. Construct  $\psi \in U^0$  and  $\chi \in W^0$  with  $\varphi = \psi + \chi$ , without using any dimension formula.
- Explain how to modify the construction when  $V$  is infinite-dimensional. Deduce that  $(U \cap W)^0 = U^0 + W^0$  holds in every vector space.

*Hint:* extend a basis of  $U \cap W$  to a basis of  $U$  and, separately, to a basis of  $W$ .

*Solution.*

- Let  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  be a basis of  $U \cap W$  (Theorem 1.2.2, Theorem 1.6.7). Extend it to a basis  $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p)$  of  $U$  and to a basis  $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{w}_1, \dots, \mathbf{w}_q)$  of  $W$

$W$  (Theorem 1.6.5).

The combined list  $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p, \mathbf{w}_1, \dots, \mathbf{w}_q)$  is independent. Let  $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i + \sum c_j \mathbf{w}_j = \mathbf{0}$ . Then  $\sum c_j \mathbf{w}_j = -\sum a_i \mathbf{x}_i - \sum b_i \mathbf{u}_i$  lies in  $W$  and in  $U$ , hence in  $U \cap W$ , so  $\sum c_j \mathbf{w}_j = \sum d_i \mathbf{x}_i$  for some  $d_i \in F$ . Since  $(\mathbf{x}, \mathbf{w})$  is a basis of  $W$ , all  $c_j$  and  $d_i$  are 0. Then  $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i = \mathbf{0}$ , and since  $(\mathbf{x}, \mathbf{u})$  is a basis of  $U$ , all  $a_i$  and  $b_i$  are 0.

Extend the combined list to a basis of  $V$  by vectors  $\mathbf{y}_1, \dots, \mathbf{y}_r$  (Theorem 1.6.5). By Theorem 3.1.5 there is  $\psi \in V^*$  with

$$\psi(\mathbf{x}_i) = 0, \quad \psi(\mathbf{u}_i) = 0, \quad \psi(\mathbf{w}_j) = \varphi(\mathbf{w}_j), \quad \psi(\mathbf{y}_l) = 0.$$

Then  $\psi$  kills a basis of  $U$ , so  $\psi \in U^0$  (Theorem 4.2.3 (b)). Put  $\chi := \varphi - \psi$ . Then  $\chi(\mathbf{x}_i) = \varphi(\mathbf{x}_i) - 0 = 0$ , because  $\mathbf{x}_i \in U \cap W$  and  $\varphi \in (U \cap W)^0$ ; and  $\chi(\mathbf{w}_j) = \varphi(\mathbf{w}_j) - \varphi(\mathbf{w}_j) = 0$ . So  $\chi$  kills a basis of  $W$ , and  $\chi \in W^0$ . Hence  $\varphi = \psi + \chi \in U^0 + W^0$ .

- (b) Replace the finite lists by sets. By Theorem 1.8.3,  $U \cap W$  has a basis  $X$ , and by Theorem 1.8.2 it extends to a basis  $X \cup P$  of  $U$  and to a basis  $X \cup Q$  of  $W$ , with  $P \cap X = Q \cap X = \emptyset$ . First,  $P \cap Q = \emptyset$ : a vector of  $P \cap Q$  lies in  $U \cap W = \text{span } X$ , which contradicts the independence of  $X \cup P$ . So every vector of  $X \cup P \cup Q$  lies in exactly one of  $X$ ,  $P$ ,  $Q$ . A linear relation among finitely many distinct vectors of  $X \cup P \cup Q$  splits into an  $X$ -part, a  $P$ -part and a  $Q$ -part, and the independence argument of (a) applies word for word; so  $X \cup P \cup Q$  is independent. By Theorem 1.8.2 it extends to a basis  $B$  of  $V$ . Every  $\mathbf{v} \in V$  is a finite combination  $\mathbf{v} = \sum_{\mathbf{b}} c_{\mathbf{b}} \mathbf{b}$  of distinct vectors of  $B$ , with unique coefficients because  $B$  is independent. Define

$$\psi(\mathbf{v}) := \sum_{\mathbf{b} \in Q} c_{\mathbf{b}} \varphi(\mathbf{b}),$$

a finite sum over the vectors of  $Q$  that occur. Adding or scaling expansions gives expansions, so by uniqueness the coefficients add and scale, and  $\psi$  is linear. It vanishes on  $X \cup P$  and agrees with  $\varphi$  on  $Q$ , so the rest of (a) is unchanged:  $\psi \in U^0$  and  $\chi = \varphi - \psi$  kills the basis  $X \cup Q$  of  $W$ . Together with the inclusion ( $\supseteq$ ) of Theorem 4.2.3 (d), which used no finiteness, this gives  $(U \cap W)^0 = U^0 + W^0$  in every vector space.

### Exercise C2: Equations of a double root.

Let  $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = 0 \text{ and } p'(1) = 0\}$ , and let  $\varepsilon_1(p) = p(1)$  and  $\eta_1(p) = p'(1)$ . Prove that  $U^0 = \text{span}(\varepsilon_1, \eta_1)$ . Deduce that  $\psi(p) = p(0) - p(2) + 2p'(1)$  does **not** vanish on  $U$ .

*Solution.* Both  $\varepsilon_1$  and  $\eta_1$  are linear (evaluation, and differentiation followed by evaluation), and they vanish on  $U$  by definition. *Independent:* if  $a\varepsilon_1 + b\eta_1 = 0$ , applying it to 1 gives  $a = 0$ , and applying it to  $x - 1$  gives  $b = 0$ . *Dimension of  $U$ :* the linear map  $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$ ,  $p \mapsto (p(1), p'(1))$ , has kernel  $U$ , and  $T(1) = (1, 0)$ ,  $T(x - 1) = (0, 1)$ , so  $T$  is surjective. By Theorem 3.3.1,  $\dim U = 4 - 2 = 2$ , and by Theorem 4.2.2,  $\dim U^0 = 2$ . The 2-dimensional subspace  $\text{span}(\varepsilon_1, \eta_1)$  of  $U^0$  is therefore all of  $U^0$  (Theorem 1.6.8).

For the deduction, suppose  $\psi \in U^0$ . Then  $\psi = a\varepsilon_1 + b\eta_1$ . Applying both sides to 1 gives  $1 - 1 + 0 = a$ , so  $a = 0$ ; applying them to  $x - 1$  gives  $-1 - 1 + 2 = b$ , so

$b = 0$ . Then  $\psi = 0$ . But  $\psi(x^2) = 0 - 4 + 4 = 0$  and  $\psi(x^3) = 0 - 8 + 6 = -2 \neq 0$ , a contradiction. Hence  $\psi \notin U^0$ . Check with an explicit witness:  $p = x(x-1)^2 \in U$  has  $\psi(p) = 0 - 2 + 0 = -2 \neq 0$ .

**Exercise C3: Dual of a direct sum decomposition.**

- (a) Let  $V$  be finite-dimensional with  $V = U \oplus W$ . Prove that  $V^* = U^0 \oplus W^0$ .
- (b) In  $\mathbb{R}^2$ , let  $U = \text{span}((1,0))$  and  $W = \text{span}((1,1))$ . Write  $\psi(x,y) = x$  as  $\psi_U + \psi_W$  with  $\psi_U \in U^0$  and  $\psi_W \in W^0$ .

*Solution.*

- (a) Since the sum is direct,  $U \cap W = \{\mathbf{0}\}$  (Theorem 1.7.4), and  $U + W = V$ . By Theorem 4.2.3 (c) and Example (First Annihilators) (a),  $U^0 \cap W^0 = (U + W)^0 = V^0 = \{\mathbf{0}\}$ . By Theorem 4.2.3 (d),  $U^0 + W^0 = (U \cap W)^0 = \{\mathbf{0}\}^0 = V^*$ . So  $V^* = U^0 + W^0$ , and the sum is direct by Theorem 1.7.4.
- (b) A functional  $ax + by$  vanishes on  $(1,0)$  if and only if  $a = 0$ , so  $U^0 = \text{span}(y)$ . It vanishes on  $(1,1)$  if and only if  $b = -a$ , so  $W^0 = \text{span}(x - y)$ . Solving  $x = sy + t(x - y)$  gives  $t = 1$  and  $s - t = 0$ , so  $s = 1$ :

$$\psi = \underbrace{y}_{\in U^0} + \underbrace{(x - y)}_{\in W^0}.$$

By (a) this decomposition is unique.